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### SEALING DEVICE FOR CAPSULE PRODUCT AND OPERATION METHOD THEREOF

#### Abstract

A sealing device for a capsule product is provided. The sealing device for the capsule product includes a base and an upper cover. The base defines an accommodating groove having an upper opening, and the base includes the positioning baffle disposed at one side of the accommodating groove. The upper cover is detachably connected to the base, a pressing portion is disposed on a lower surface of the upper cover.

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## Background/Summary

### TECHNICAL FIELD

[0001] The present disclosure relates to a technical field of sealing apparatus, and in particular to a sealing device for a capsule product and an operation method thereof.

### BACKGROUND

[0002] Capsule products each is typically composed of a capsule shell and a product contained therein. A corresponding capsule shell includes a cup body and a sealing cover, the cup body features a protruding edge at a top end thereof for tightly sealing with the sealing cover after a desired product, such as coffee, barley, milk powder, tea, etc., has been filled in the cup body, so as to enable sealing and storage of the desired product. When using the capsule products, the user only needs to puncture or uncover a corresponding sealing cover to get in contact with a corresponding desired product contained in a corresponding cup body and use the corresponding desired product. However, most of the capsule products currently in the market are pre-made products, making it difficult to meet different customer requirements. Moreover, although the corresponding capsule shell of the capsule products remain intact during use, it is often discarded after use, undoubtedly representing a waste of resources.

### SUMMARY

[0003] Embodiments of the present disclosure provide a sealing device for a capsule product and an operation method thereof, which improves sealing stability of a sealing piece and a cup body through a positioning baffle and is capable of customizing the capsule product according to different customer requirements.

[0004] In a first aspect, the embodiments of the present disclosure provide the sealing device for the capsule product, including a base and an upper cover. The base defines an accommodating groove having an upper opening, and the base includes the positioning baffle disposed at one side of the accommodating groove. The upper cover is detachably connected to the base, a pressing portion is disposed on a lower surface of the upper cover.

[0005] In some embodiments, an annular protrusion is disposed on an upper surface of the base and is annularly disposed at an outer periphery of the upper opening of the accommodating groove.

[0006] In some embodiments, a positioning concave portion is further defined on the lower surface of the upper cover, the positioning baffle is at least partially inserted into the positioning concave portion and slides in the positioning concave portion.

[0007] In some embodiments, the positioning concave portion is of an annular structure.

[0008] In some embodiments, the capsule product includes a capsule shell and a sealing piece, the capsule shell includes a cup body having a receiving groove, a flange connected to a top portion of the cup body and extending in a radial direction, and a crimping edge connected to an outer periphery of the flange, the sealing piece is attached to the flange and is configured to seal the receiving groove. An annular limiting portion is further disposed on the lower surface of the upper cover and is annularly disposed at an outer periphery of the pressing portion, an annular recess is formed between the annular limiting portion and the pressing portion, the cup body is disposed in the accommodating groove, the flange is clamped between the pressing portion and the annular protrusion, the crimping edge is inserted into the annular recess, and an outer periphery of the crimping edge abuts against an inner wall surface of the annular limiting portion.

[0009] In some embodiments, the upper cover includes a cover body and a top cover, the pressing portion is disposed on a bottom portion of the cover body, the pressing portion is of an annular structure matched with the annular protrusion, the cover body defines a through hole at an inner side of the pressing portion, the top cover is detachably connected to a top portion of the cover body for covering or exposing the through hole. When the cover body is disposed on the base and

the top cover is disconnected to the top portion of the cover body, a powdered product is placed into the receiving groove through the through hole.

[0010] In some embodiments, a protruding structure is disposed on an upper surface of the upper cover, a shape of the protruding structure is matched with a shape of the upper opening of the accommodating groove, and the protruding structure is configured to flatten a powdered product in the receiving groove.

[0011] In some embodiments, in a radial direction, a distance between of an inner wall surface of the positioning baffle and an outer wall surface of the annular protrusion is not less than 0.1 mm and is not greater than 5 mm.

[0012] In some embodiments, in the radial direction, a dimension of the annular protrusion is not less than 0.2 mm and not greater than 5 mm; and/or in a vertical direction, the dimension of the annular protrusion is not less than 0.1 mm and not greater than 10 mm.

[0013] In some embodiments, the pressing portion is a planar structure extending in a horizontal direction.

[0014] In some embodiments, the base defines a positioning groove at an outer periphery thereof, a positioning protruding portion is disposed at an outer periphery of the upper cover, the positioning groove is configured to limit the positioning protruding portion and is matched with the positioning protruding portion, so as to prevent the upper cover and the base from being separated from each other in the radial direction.

[0015] In some embodiments, the positioning baffle is of an arc-shaped structure, the base has a circle passing through the positioning baffle in a circumferential direction thereof, an arc length occupied by the positioning baffle on the circle does not exceed a circumference of the circle and is not less than one quarter of the circumference of the circle.

[0016] In some embodiments, an anti-slip pad is disposed on a lower surface of the base.

[0017] In some embodiments, the sealing device for the capsule product further includes a cutting structure, the cutting structure includes a mounting base and a blade assembly, a limiting groove is defined on a first side of the mounting base, the blade assembly includes a fixing base and a blade fixed on the fixing base, the mounting base includes a limiting hole penetrating through two opposite sides thereof, and the limiting hole is defined in the limiting groove, the blade partially extends out from the limiting hole.

[0018] An outer periphery of the capsule shell abuts against a groove side wall of the limiting groove, the capsule shell rotates in the limiting groove, the blade is configured to cut a part of the sealing piece when the capsule shell rotates with respect to the mounting base, where the part of the sealing piece is disposed at an inner side of the flange.

[0019] In some embodiments, an included angle is formed between the blade and a horizontal direction, the blade partially extends out from the limiting hole and is inclined in a direction away from an edge of the capsule shell.

[0020] In some embodiments, the included angle between the blade and the horizontal direction is  $\alpha$ , the  $\alpha$  satisfies a relationship as follows:

$$55^{\circ} \leq \alpha \leq 95^{\circ}.$$

[0021] In some embodiments, the fixing base includes a first base body and a second base body, the first base body is detachably connected to the second base body, the first base body includes a first mounting surface, the first mounting surface is inclined, the second base body includes a second mounting surface, the second mounting surface is inclined, the blade is clamped between the first mounting surface and the second mounting, so as to form the included angle with the horizontal direction.

[0022] In some embodiments, a first mounting groove and a second mounting groove are defined on a second side of the mounting base facing away from the limiting groove, the first mounting groove is communicated with the limiting hole, the blade assembly is fixed at the first mounting

groove in a use state, and the blade assembly is accommodated in the second mounting groove in an idle state.

[0023] In some embodiments, the mounting base includes a mounting portion, the mounting portion defines the first mounting groove, the mounting portion includes a hook, the fixing base defines a clamping groove. When the fixing base is fixed at the first mounting groove, the hook is snap-fitted to the clamping groove.

[0024] In a second aspect, the embodiments of the present disclosure further provides a first embodiment of the operation method of the sealing device for the capsule product, in which the capsule product includes the capsule shell and the powdered product contained in the capsule shell, the capsule shell includes the cup body and the sealing piece, the cup body defines the receiving groove and includes the flange at the top portion thereof, the flange extends in the radial direction, the sealing piece is attached to the flange and is configured to seal the receiving groove, the sealing device for the capsule product includes the base and the upper cover, the base defines the accommodating groove having the upper opening, the annular protrusion is disposed on the upper surface of the base and is annularly disposed at the outer periphery of the upper opening of the accommodating groove, the accommodating groove is configured to accommodate the cup body, the annular protrusion is configured to support the flange, the base includes the positioning baffle disposed at the one side of the accommodating groove, and the positioning baffle is configured to limit the sealing piece, the upper cover is detachably connected to the base, the pressing portion is disposed on the lower surface of the upper cover.

[0025] The operation method of the sealing device for the capsule product includes following steps.

[0026] Placing the cup body of the capsule shell of the capsule product in the accommodating groove of the base of the sealing device for the capsule product, supporting the flange of the cup body on the annular protrusion of the base.

[0027] Placing the powdered product into the receiving groove of the cup body.

[0028] Abutting an outer wall surface of the sealing piece of the capsule shell against an inner wall of the positioning baffle of the base, and attaching the sealing piece to the flange.

[0029] Covering the base by the upper cover of the sealing device of the capsule product, so as to seal and attach the sealing piece to the flange.

[0030] In a third aspect, the embodiments of the present disclosure further provides a second embodiment of the operation method of the sealing device for the capsule product, in which the capsule product includes the capsule shell and the powdered product contained in the capsule shell, the capsule shell includes the cup body and the sealing piece, the cup body defines the receiving groove and includes the flange at the top portion thereof, the flange extends in the radial direction, the sealing piece is attached to the flange and is configured to seal the receiving groove, the sealing device for the capsule product includes the base and the upper cover, the base defines the accommodating groove having the upper opening, the annular protrusion is disposed on the upper surface of the base and is annularly disposed at the outer periphery of the upper opening of the accommodating groove, the accommodating groove is configured to accommodate the cup body, the annular protrusion is configured to support the flange, the base includes the positioning baffle disposed at the one side of the accommodating groove, and the positioning baffle is configured to limit the sealing piece, the upper cover includes the cover body and the top cover, the cover body is detachably connected to the base, and the pressing portion is disposed on the bottom portion of the cover body, the cover body defines the through hole at the inner side of the pressing portion, the top cover is detachably connected to the top portion of the cover body for covering or exposing the through hole. The operation method of the sealing device for the capsule product includes following steps.

[0031] Placing the cup body of the capsule shell of the capsule product in the accommodating groove of the base of the sealing device for the capsule product, supporting the flange of the cup body on the annular protrusion of the base.

[0032] Placing the powdered product into the receiving groove of the cup body, or covering the base by the cover body of the upper cover of the sealing device for the capsule product and placing the powdered product into the receiving groove through the through hole of the cover body.

[0033] Removing the cover body, abutting the outer wall surface of the sealing piece of the capsule shell against the inner wall of the positioning baffle of the base, and attaching the sealing piece to the flange.

[0034] Covering the base by the upper cover of the sealing device of the capsule product, so as to seal and attach the sealing piece to the flange.

[0035] According to the sealing device for the capsule product of the embodiments of the present disclosure, the upper cover is detachably connected to the base, and the annular protrusion is configured to support the flange, in a process of covering the base by the upper cover, the annular protrusion is matched with the pressing portion of the upper cover to clamp the sealing piece and the flange, thereby tightly attaching the sealing piece to the flange, in this way, external air, moisture, or contaminants are effectively prevented from entering an interior of the capsule shell, thereby enabling freshness and quality of a solid or semi-solid product, such as coffee, barley, milk powder, tea, etc., contained in the capsule shell. When the sealing pieces covers the cup body, the positioning baffle is configured to limit the sealing piece to prevent the sealing piece from deviating, thereby improving a sealing effect.

[0036] The sealing device for the capsule product of the embodiments of the present disclosure adopts a simple mechanical structure, which does not need complex processing or expensive materials, the base and the upper cover are both simple in structures, easy to produce, and have low production costs, thereby having a better market competitiveness. At the same time, since the sealing device for the capsule product is easy to manufacture, a use cost of a user is also reduced. Moreover, the sealing device for the capsule product of the present disclosure is small in size, which is beneficial to saving storage space and is convenient to carry and transport.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0037] In order to more clearly illustrate embodiments of the present disclosure or technical solutions in the prior art, accompanying drawings required in description of the embodiments or the prior art are briefly described below, and obviously, the accompanying drawings in the following description are merely some embodiments of the present disclosure, and for those who skilled in the art, other drawings may be obtained according to structures shown in these drawings without creative efforts.

[0038] FIG. 1 is a structural schematic diagram of a capsule product according to one embodiment of the present disclosure.

[0039] FIG. 2 is a structural schematic diagram of a sealing device for the capsule product according to one embodiment of the present disclosure.

[0040] FIG. 3 is an exploded schematic diagram of the sealing device for the capsule product according to one embodiment of the present disclosure.

[0041] FIG. 4 is a cross-sectional schematic diagram of the sealing device for the capsule product according to one embodiment of the present disclosure.

[0042] FIG. 5 is a cross-sectional schematic diagram of the sealing device for the capsule product according to one embodiment of the present disclosure where the sealing device for the capsule product is exploded.

[0043] FIG. 6 is a top schematic diagram of a base according to one embodiment of the present disclosure.

[0044] FIG. 7 is a cross-sectional schematic diagram of the sealing device for the capsule product

according to another embodiment of the present disclosure.

[0045] FIG. **8** is an exploded schematic diagram of the sealing device for the capsule product according to another embodiment of the present disclosure.

[0046] FIG. **9** is a structural schematic diagram of a cutting structure according to one embodiment of the present disclosure.

[0047] FIG. **10** is a structural schematic diagram of a blade assembly according to one embodiment of the present disclosure where the blade assembly is in a use state.

[0048] FIG. **11** is a structural schematic diagram of the blade assembly according to one embodiment of the present disclosure where the blade assembly is in an idle state.

[0049] FIG. **12** is a structural schematic diagram of the blade assembly according to one embodiment of the present disclosure.

[0050] FIG. **13** is a structural schematic diagram of the cutting structure according to one embodiment of the present disclosure where the cutting structure is in a use state.

[0051] FIG. **14** is a schematic flowchart of an operation method of the sealing device for the capsule product according to one embodiment of the present disclosure.

[0052] FIG. **15** is a schematic flowchart of the operation method of the sealing device for the capsule product according to another embodiment of the present disclosure

[0053] Reference numerals in the drawings: **1**. sealing device; **10**. base; **11**. accommodating groove; **12**. annular protrusion; **13**. positioning baffle; **14**. positioning groove; **15**. anti-slip pad; **20**. upper cover; **21**. cover body; **22**. top cover; **23**. pressing portion; **24**. positioning concave portion; **25**. annular limiting portion; **26**. annular recess; **27**. protruding structure; **28**. through hole; **29**. positioning protruding portion; **30**. cutting structure; **31**. mounting base; **311**. limiting groove; **312**. limiting hole; **313**. first mounting groove; **314**. second mounting groove; **315**. mounting portion; **316**. hook; **32**. blade assembly; **321**. fixing base; **3211**. first base body; **3212**. second base body; **3215**. clamping groove; **322**. blade; **200**. capsule shell; **200A**. receiving groove; **210**. cup body; **220**. flange; **230**. crimping edge; **240**. sealing piece.

[0054] Implementations, functional features, and advantages of the present disclosure are further described with reference to the accompanying drawings in combination with the embodiments.

#### DETAILED DESCRIPTION OF EMBODIMENTS

[0055] In order to better understand technical solutions of the present disclosure, embodiments of the present disclosure are described in detail below with reference to accompanying drawings.

[0056] When following description refers to the accompanying drawings, unless otherwise indicated, the same numbers in different drawings represent the same or similar elements.

Implementations described in following exemplary embodiments do not represent all implementations consistent with this application. Instead, they are merely examples of apparatuses and methods consistent with some aspects of the present disclosure as recited in appended claims.

[0057] In the description of the present disclosure, it should be understood that terms “first”, “second”, etc. are only used for descriptive purposes, and cannot be understood as indicating or implying relative importance. Those who skilled in the art would understand specific meanings of the foregoing terms in the present disclosure according to specific situations. In addition, in the description of the present disclosure, unless otherwise stated, “a plurality of” means two or more, “and/or” used herein is merely an association relationship describing associated objects. For example, A and/or B may indicate three cases that A exists alone, A and B exist, and B exists alone, and a character “/” in the specification generally indicates an “or” relationship between the associated objects

[0058] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those who skilled in the art to which the present disclosure belongs. The terms used in the specification are only for the purpose of describing specific embodiments, and are not intended to limit the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of associated listed items.

[0059] The embodiments of the present disclosure provides a sealing device **1** for a capsule product, the sealing device **1** for the capsule product is configured to seal the capsule product, the capsule product includes a capsule shell **200** and a solid or semi-solid product contained in the capsule shell **200**, please refer to FIG. **1**, the capsule shell **200** includes a cup body **210**, flange **210**, and a sealing piece **240**, the cup body **210** defines the receiving groove **200A**, the flange **220** is annularly connected to a top portion of the cup body **210** and extends in a radial direction, and the sealing piece **240** is attached to the flange **220** and is configured to seal the receiving groove **200A**. Specifically, the solid or semi-solid product contained in the capsule shell **200** is selected from granular, pasty, powdered, or leaf-like products, such as coffee, barley, milk powder, tea, etc. For example, the capsule product of the present disclosure is capsule coffee.

[0060] Please refer to FIGS. **2-3**, the sealing device **1** for the capsule product, includes a base **10** and an upper cover **20**. The base **10** defines an accommodating groove **11** having an upper opening, the accommodating groove **11** is configured to accommodate the cup body **210**, an annular protrusion **12** is disposed on an upper surface of the base **10** and is annularly disposed at an outer periphery of the upper opening of the accommodating groove **11**, the annular protrusion **12** is configured to support the flange **220**. A shape of the accommodating groove **11** is matched with a shape of the cup body **210**, an outer surface of the cup body **210** is attached to a groove wall of the accommodating groove **11**, so as to stability of the cup body **210** fixing in the accommodating groove **11**. In some embodiments, the accommodating groove **11** is of a hemispherical structure. In some other embodiments, the accommodating groove **11** is selected from a columnar structure, a circular truncated cone structure, etc. Certainly, the accommodating groove **11** is capable of accommodating the cup body **210** shaped in different structures, in some other embodiments, the cup body **210** is also not attached to the groove wall of the accommodating groove **11**, but is only supported on the annular protrusion **12** by the flange **220** to fix with respect to the base **10**. The base **10** includes a positioning baffle **13** disposed at one side of the accommodating groove **11**.

[0061] The upper cover **20** is detachably connected to the base **10**, please refer to FIGS. **4-5**, a pressing portion **23** is disposed on a lower surface of the upper cover **20**, when the upper cover **20** covers the base **10**, the pressing portion **23** is tightly attached to the annular protrusion **12** to seal and attach the sealing piece **240** to the flange **220**. Specifically, when using the sealing device **1** for the capsule product of the present disclosure, the cup body **210** is first placed in the accommodating groove **11** and is supported on the annular protrusion **12** by the flange **220**, then the solid or semi-solid product is placed in the receiving groove **200A** of the cup body **210**, an outer wall surface of the sealing piece **240** is abutted against an inner wall of the positioning baffle **13**, and then is attached to the flange **220**, the upper cover **20** is provided to cover the base **10**, the annular protrusion **12** and the pressing portion **23** are matched in a vertical direction to seal and attach the sealing piece **240** to the flange **220**, and finally, the upper cover **20** is detached from the base **10** to obtain a new capsule product. It may be understood that an adhesive layer is disposed on a bottom surface of the sealing piece **240**, and after a pressure is applied to the sealing piece **240** and the flange **220**, the adhesive layer is configured to attach and fix the sealing piece **240** and the flange **220**. In this way, external air, moisture, or contaminants are effectively prevented from entering an interior of the capsule shell **200**, thereby enabling freshness and quality of the solid or semi-solid product, such as coffee, barley, milk powder, tea, etc., contained in the capsule shell **200**.

[0062] According to the sealing device **1** for the capsule product of the embodiments of the present disclosure, the upper cover **20** is detachably connected to the base **10**, and the annular protrusion **12** is configured to support the flange **220**, in a process of covering the base **10** by the upper cover **20**, the annular protrusion **12** is matched with the pressing portion **23** of the upper cover **20** to clamp the sealing piece **240** and the flange **220**, thereby tightly attaching the sealing piece **240** to the flange **220**. When the sealing piece **240** covers the cup body **210**, the positioning baffle **13** is configured to limit the sealing piece **240** to prevent the sealing piece **240** from deviating, thereby

improving a sealing effect of the sealing piece **240**. The sealing device **1** for the capsule product of the embodiments of the present disclosure adopts a simple mechanical structure, which does not need complex processing or expensive materials, the base **10** and the upper cover **20** are both formed by plastic injection molding, being simple in structures, easy to produce, and having low production costs, thereby enabling a better market competitiveness. At the same time, since the sealing device **1** for the capsule product is easy to manufacture, a use cost of a user is also reduced. Moreover, the sealing device **1** for the capsule product of the present disclosure is small in size, which is beneficial to saving storage space and is convenient to carry and transport.

[0063] It may be understood that, in an operation process of a capsule coffee maker, the capsule coffee maker may puncture the sealing piece **240**, while the cup body **210** is intact, the present disclosure allows the capsule shell **200** to be cleaned and reused once used, the user may select a desired type of coffee to re-place in the cup body **210**, and then the upper cover **20** and the base **10** are matched to attach a new sealing piece **240** to the cup body **210** to obtain a new capsule coffee. The present enables a reuse of the cup body **210** of the capsule shell **200**, which helps to reduce generation of waste and is more environmentally friendly. The user may also freely select types, brands, and tastes of coffee powder according to their personal preferences and taste requirements, and even try to mix different kinds of coffee powder to create a unique coffee flavor, thereby increasing fun and satisfaction of use.

[0064] In some embodiments, please refer to FIG. 5, a positioning concave portion **24** is further defined on the lower surface of the upper cover **20**, the positioning baffle **13** is at least partially inserted into the positioning concave portion **24** and slides in the positioning concave portion **24**. It should be noted that, the upper cover **20** and the base **10** are rotated to improve pressure applied on the flange **220** and the sealing piece **240** by the annular protrusion **12** and the pressing portion **23**, thereby improve sealing stability, and at least partially inserting the positioning baffle **13** into the positioning concave portion **24** and sliding the positioning baffle **13** in the positioning concave portion **24** have a limitation on rotation of the upper cover **20** and the base **10**, in this way, the upper cover **20** are prevented from deviating or shaking during the rotation, and the upper cover **20** is further prevented from being disengaged in the radial direction when rotating with respect to the base **10**, which further improves the sealing stability.

[0065] When the upper cover **20** rotates at a certain angle (less than 360°) with respect to the base **10**, the positioning concave portion **24** is of an arc-shaped structure in a circumferential direction of the upper cover **20**; when the upper cover **20** rotates 360° with respect to the base **10**, the positioning concave portion **24** is of an annular structure in the circumferential direction of the upper cover **20**. The positioning baffle **13** may adapt to a shape of the positioning concave portion **24** to be of the annular structure, may also be of the arc-shaped structure or a cylindrical structure. Specific shapes of the positioning baffle **13** and the positioning concave portion **24** are not limited in the present disclosure.

[0066] In some embodiments, the capsule shell **200** further includes a crimping edge **230** connected to an outer periphery of the flange **220**. It may be understood that, in order to prevent an outer periphery of the capsule shell **200** from damaging humans' hands, the crimping edge **230** is provided to connect to the outer periphery of the flange **220**, and if the crimping edge **230** is flattened in the process of covering the base **10** by the upper cover **20**, a protection effect of the crimping edge **230** is reduced. Please further refer to FIG. 5, an annular limiting portion **25** is further disposed on the lower surface of the upper cover **20** and is annularly disposed at an outer periphery of the pressing portion **23**, an annular recess **26** is formed between the annular limiting portion **25** and the pressing portion **23**, the flange **220** is clamped between the pressing portion **23** and the annular protrusion **12**, the crimping edge **230** is inserted into the annular recess **26**, and an outer periphery of the crimping edge **230** abuts against an inner wall surface of the annular limiting portion **25**. The annular recess **26** provides a safe space for the crimping edge **230**, so that in the process of covering the base **10** by the upper cover **20**, the crimping edge **230** is avoided from



being flattened or deformed, which is beneficial for the crimping edge **230** to maintain an original shape and size, thereby ensuring that the protection effect of the crimping edge **230** does not decrease. The outer periphery of the crimping edge **230** abuts against the inner wall surface of the annular limiting portion **25**, so that stability of the capsule shell **200** between the base **10** and the upper cover **20** is improved, and the capsule shell **200** is further avoided from shifting, etc.

[0067] In some embodiments, please refer to FIG. 6, in the radial direction, a dimension **d1** of the annular protrusion **12** is not less than 0.2 mm and not greater than 5 mm. The dimension **d1** of the annular protrusion **12** is matched with a dimension of the flange **220**, that is, the dimension of the flange **220** is not less than 0.2 mm and not greater than 5 mm, in this way, increasing a contact area between the flange **220** and the sealing piece **240** is capable of improving a sealing effect of the capsule product. Please further refer to FIG. 4, in a vertical direction, the dimension **h** of the annular protrusion **12** is not less than 0.1 mm and not greater than 10 mm. Specifically, in the vertical direction, when the dimension **h** of the annular protrusion **12** is less than 0.1 mm, pressing the upper cover **20** on the base **10** may squeeze the crimping edge **230**, and when the dimension **h** of the annular protrusion **12** is greater than 10 mm, a height of the positioning baffle **13** is affected, it should be understood that, the positioning baffle **13** is capable of limiting the sealing piece **240** only if the height of the positioning baffle **13** is higher than a height of the annular protrusion **12**.

[0068] In some embodiments, please refer to FIG. 6, the positioning baffle **13** is of an arc-shaped structure, the base **10** has a circle **O** passing through the positioning baffle **13** in a circumferential direction thereof, an arc length occupied by the positioning baffle **13** on the circle **O** does not exceed a circumference of the circle and is not less than one quarter of the circumference of the circle **O**. That is, the positioning baffle **13** is not of an annular structure and has a notch on one side thereof, in this way, when the user holds one side of the sealing piece **240**, enough operation space is obtained at the notch, so that the sealing piece **240** is easily and flatly attached to the flange **220**. The arc length occupied by the positioning baffle **13** on the circle **O** does not exceed the circumference of the circle and is not less than one quarter of the circumference of the circle **O**, and an arc length of the positioning baffle **13** is greater than or equal to one quarter of the circumference of the circle **O**, so that a contact area between the positioning baffle **13** and the sealing piece **240** is ensured, thereby ensuring a positioning effect of the sealing piece **240**.

[0069] Certainly, in some embodiments, the positioning baffle **13** includes a plurality of sub-baffles, the plurality of the sub-baffles are disposed at intervals and jointly form a section of the arc-shaped structure, and a length of the arc-shaped structure in the embodiments, that is, an arc length between two outermost sub-baffles of the plurality of the sub-baffles, is set with reference to the above-mentioned form. In the embodiments of the present disclosure, the positioning baffle **23** may be either a continuous structure or a discontinuous structure, and both the continuous structure and the discontinuous structure enables positioning of the sealing piece **240**.

[0070] In some embodiments, please refer to FIG. 6, in the radial direction, a distance **d2** between of an inner wall surface of the positioning baffle **13** and an outer wall surface of the annular protrusion **12** is not less than 0.1 mm and is not greater than 5 mm. That is, in the radial direction, a dimension of the annular recess **26** is not less than 0.1 mm and not greater than 0.5 mm, the dimension of the annular recess **26** is matched with a dimension of the crimping edge **230**; if the dimension of the crimping edge **230** is less than 0.1 mm, it is difficult to ensure structural strength thereof, and a processing difficulty of the crimping edge **230** is increased; and if the dimension of the crimping edge **230** is greater than 0.5 mm, a size of the capsule product is affected, and a transportation cost of the capsule product is further increased.

[0071] Please refer to FIG. 2, in some embodiments, an anti-slip pad **15** is disposed on a lower surface of the base **10**. The anti-skid pad **15** is capable of increasing a friction force between the base **10** and a table or a countertop, which is beneficial for the user to seal the flange **220** and the sealing piece **240** by pressing and rotating the upper cover **20**.

[0072] Please refer to FIGS. 2-3, in some embodiments, a protruding structure **27** is disposed on an

upper surface of the upper cover **20**, a shape of the protruding structure **27** is matched with a shape of the upper opening of the accommodating groove **11**, and the protruding structure **27** is configured to flatten a powered product in the receiving groove **200A**. The protruding structure **27** flattens the powered product in the receiving groove **200A** to ensure that a surface of the powered product is flat and free of recesses or protrusions, thereby preventing the powered product in the receiving groove **200A** from excessively protruding and affecting the sealing effect of the sealing piece **240**.

[0073] In some embodiments, the pressing portion **23** is a planar structure extending in a horizontal direction. The pressing portion **23** of the planar structure provides a uniform pressing force when cooperating with the protruding structure **27** to ensure that the sealing piece **240** is uniformly clamped on the flange **220**.

[0074] Please refer to FIGS. 7-8, in some embodiments, the upper cover **20** includes a cover body **21** and a top cover **22**, the pressing portion **23** is disposed on a bottom portion of the cover body **21**, the pressing portion **23** is of an annular structure matched with the annular protrusion **12**, the cover body **21** defines a through hole **28** at an inner side of the pressing portion **23**, the top cover **22** is detachably connected to a top portion of the cover body **21** for covering or exposing the through hole **28**. When the cover body **21** covers the base **10** and the top cover **22** is detached from the top portion of the cover body **21**, a powdered product is placed into a receiving groove **200A** through the through hole **28**, in this way, a product filling process is simpler, more convenient, and faster, the powdered product, such as the coffee powder, is prevented from being spilling onto the flange **220**, which avoids the need to clean the flange **220**, saves time, further improves operational convenience.

[0075] Please refer to FIG. 5, in some embodiments, the base **10** defines a positioning groove **14** at an outer periphery thereof, a positioning protruding portion **29** is disposed at an outer periphery of the upper cover **20**, the positioning groove **14** is configured to limit the positioning protruding portion **29** and is matched with the positioning protruding portion **29**, so as to prevent the upper cover **20** and the base **10** from being separated from each other in the radial direction, in this way, stability when the upper cover **20** rotates with respect to the base **10** is improved.

[0076] Please refer to FIGS. 9-12, in some embodiments, the sealing device **1** for the capsule product further includes a cutting structure **30**, the cutting structure **30** includes a mounting base **31** and a blade assembly **32**, a limiting groove **311** is defined on a first side of the mounting base **31**, the blade assembly **32** includes a fixing base **321** and a blade **322** fixed on the fixing base **321**, the mounting base **31** includes a limiting hole **312** penetrating through two opposite sides thereof, and the limiting hole **312** is defined in the limiting groove **311**, the blade **322** partially extends out from the limiting hole **312**. The outer periphery of the capsule shell **200** abuts against a groove side wall of the limiting groove **311**, the capsule shell **200** rotates in the limiting groove **311**, the blade **322** is configured to cut a part of the sealing piece **240** when the capsule shell **200** rotates with respect to the mounting base **31**, where the part of the sealing piece **240** is disposed at an inner side of the flange **220**. The blade assembly **32** is mounted on the mounting base **31** through the fixing base **321**, and the blade **322** partially extends out from the limiting hole **312**. When the outer periphery of the capsule shell **200** abuts against the groove side wall of the limiting groove **311** and the capsule shell **200** rotates, the blade **322** is configured to accurately cut the part of sealing piece **240** disposed at the inner side of the flange **220**, so that the sealing piece **240** originally with good sealing is effectively and accurately cut, which avoids damage to the capsule shell **200** due to improper cutting and also avoids residues of the sealing piece **240** due to the improper cutting. The limiting groove **311** ensures that the capsule shell **200** is relatively stable in a rotating process and is not prone to slipping or deviating, improves stability and safety of the blade **322** in a cutting process, and further improves cutting efficiency of the blade **322** at the same time.

[0077] Please refer to FIG. 12, an included angle is formed between the blade **322** and the horizontal direction, the blade **322** partially extends out from the limiting hole **312** and is inclined

in a direction away from an edge of the capsule shell **200**. The included angle is formed between the blade **322** and the horizontal direction, so that the blade **322** is capable of generating a sharper cutting angle when cutting the capsule shell **200**, so that the blade **322** is easier to cut into and cut the sealing piece **240**, cutting efficiency and accuracy thereof are improved, and a risk that the capsule shell **200** may be stuck or damaged in the cutting process is reduced. The blade **322** is inclined in the direction away from the edge of the capsule shell **200**, so that frictional resistance between the blade **322** and the capsule shell **200** during the cutting process is reduced. The included angle between the blade **322** and the horizontal direction is  $\alpha$ , the  $\alpha$  satisfies a relationship as follows:

$55^{\circ} \leq \alpha \leq 95^{\circ}$ .

[0078] Specifically, the  $\alpha$  is selected from  $55^{\circ}$ ,  $60^{\circ}$ ,  $65^{\circ}$ ,  $70^{\circ}$ ,  $75^{\circ}$ ,  $80^{\circ}$ ,  $85^{\circ}$ ,  $90^{\circ}$ ,  $95^{\circ}$ , etc.

[0079] The fixing base **321** includes a first base body **3211** and a second base body **3212**, the first base body **3211** is detachably connected to the second base body **3212**, the first base body **3211** includes a first mounting surface, the first mounting surface is inclined, the second base body **3212** includes a second mounting surface, the second mounting surface is inclined, the blade **322** is clamped between the first mounting surface and the second mounting, so as to form the included angle with the horizontal direction. Since the first base body **3211** and the second base body **3212** are detachably connected, the user may easily separate the first base body **3211** and the second base body **3212**, which is convenient for installing or replacing the blade **322**. Moreover, providing the first mounting surface on the first base body **3211** and providing the second mounting surface on the second base body **3212** facilitates inclination of the blade **322**.

[0080] Please refer to FIGS. **10-11**, a first mounting groove **313** and a second mounting groove **314** are defined on a second side of the mounting base **31** facing away from the limiting groove **311**, the first mounting groove **313** is communicated with the limiting hole **312**, the blade assembly **32** is fixed at the first mounting groove **313** in a use state, and the blade assembly **32** is accommodated in the second mounting groove **314** in an idle state. In the use state, the blade assembly **32** is firmly fixed at the first mounting groove **313** to ensure stability and accuracy of the cutting process; and in the idle state, the blade assembly **32** is accommodated in the second mounting groove **314**, thereby avoiding a safety risk that may be caused by exposure of the blade **322**, and also saving space. The blade assembly **32** is easily switched between the first mounting groove **313** and the second mounting groove **314**, thereby enabling switching between storage and use.

[0081] The mounting base **31** includes a mounting portion **315**, the mounting portion **315** defines the first mounting groove **313**, the mounting portion **315** includes blocking plates, the blocking plates are connected to one side of the mounting base **31** and are enclosed to form the first mounting groove **313**, the mounting portion **315** includes a hook **316**, the fixing base **321** defines a clamping groove **3215**. When the fixing base **321** is fixed at the first mounting groove **313**, the hook **316** is snap-fitted to the clamping groove **3215**. Snap connection between the hook **316** and the clamping groove **3215** ensures stability of the fixing base **321** on the mounting portion **315**. Such connection manner is not prone to loosening, and effectively prevents the fixing base **321** from shifting or detaching during use, thereby ensuring stability and safety of the cutting process. In some embodiments, the mounting base **31** further includes partition plates, and the partition plates are disposed at intervals, or are sequentially connected to form the second mounting groove **314**.

[0082] In a second aspect, the embodiments of the present disclosure further provides a first embodiment of an operation method of the sealing device **1** for the capsule product, a structure of the capsule product is the same as that of the foregoing embodiment, and details are not described herein again, the sealing device **1** for the capsule product includes the base **10** and the upper cover **20**, the base **10** defines the accommodating groove **11** having the upper opening, the accommodating groove **11** is configured to accommodate the cup body **210**, the annular protrusion

**12** is disposed on the upper surface of the base **10** and is annularly disposed at the outer periphery of the upper opening of the accommodating groove **11**, the annular protrusion **12** is configured to support the flange **220**, the base **10** includes the positioning baffle **13** disposed at the one side of the accommodating groove **11**, the upper cover **20** is detachably connected to the base **10**, the pressing portion **23** is disposed on the lower surface of the upper cover **20**.

[0083] Taking the capsule coffee as the capsule product and referring to FIG. **14**, the operation method of the sealing device **1** for the capsule product includes following steps. [0084] **S01**: placing the cup body in the accommodating groove, supporting the flange on the annular protrusion. [0085] **S02**: placing the coffee powder into the receiving groove. [0086] **S03**: abutting the outer wall surface of the sealing piece against the inner wall of the positioning baffle, and attaching the sealing piece to the flange. [0087] **S04**: covering the base by the upper cover, so as to seal and attach the sealing piece to the flange.

[0088] In some embodiments, after the upper cover **20** covers the base **10**, the upper cover **20** is rotated to seal and attach the sealing piece **240** to the flange **220**.

[0089] In some embodiments, the protruding structure **27** is disposed on the upper surface of the upper cover **20**, the shape of the protruding structure **27** is matched with the shape of the upper opening of the accommodating groove **11**, in the **S02**, the protruding structure **27** is configured to flatten the coffee powder in the receiving groove **200A**.

[0090] In a third aspect, the embodiments of the present disclosure further provides a second embodiment of the operation method of the sealing device **1** for the capsule product, the structure of the capsule product is the same as that of the foregoing embodiment, and details are not described herein again. The sealing device **1** for the capsule product includes the base **10** and the upper cover **20**, the base **10** defines the accommodating groove **11** having the upper opening, the accommodating groove **11** is configured to accommodate the cup body **210**, the annular protrusion **12** is disposed on the upper surface of the base **10** and is annularly disposed at the outer periphery of the upper opening of the accommodating groove **11**, the annular protrusion **12** is configured to support the flange **220**, the base **10** includes the positioning baffle **13** disposed at the one side of the accommodating groove **11**, the upper cover **20** includes the cover body **21** and the top cover **22**, the cover body **21** is detachably connected to the base **10**, and the pressing portion **23** is disposed on the bottom portion of the cover body **21**, the cover body **21** defines the through hole **28** at the inner side of the pressing portion **23**, the top cover **22** is detachably connected to the top portion of the cover body **21** for covering or exposing the through hole **28**.

[0091] Taking the capsule coffee as the capsule product and referring to FIG. **15**, the operation method of the sealing device **1** for the capsule product includes following steps. [0092] **S10**: placing the cup body in the accommodating groove, supporting the flange on the annular protrusion. [0093] **S20**: placing the coffee powder into the receiving groove or covering the base by the cover body and placing the coffee powder into the receiving groove through the through hole. [0094] **S30**: removing the cover body, abutting the outer wall surface of the sealing piece against the inner wall of the positioning baffle, and attaching the sealing piece to the flange. [0095] **S40**: covering the base by the upper cover, so as to seal and attach the sealing piece to the flange.

[0096] In some embodiments, after the upper cover **20** covers the base **10**, the upper cover **20** is rotated to seal and attach the sealing piece **240** to the flange **220**.

[0097] In some embodiments, the protruding structure **27** is disposed on the upper surface of the upper cover **20**, the shape of the protruding structure **27** is matched with the shape of the upper opening of the accommodating groove **11**, in the **S20**, the protruding structure **27** is configured to flatten the coffee powder in the receiving groove **200A**.

[0098] Same or similar reference numerals in the drawings of the embodiments of the present disclosure correspond to the same or similar components. In the description of the present disclosure, it should be understood that orientations or working position relationships indicated by terms “upper”, “lower”, “left”, “right”, etc. are only for ease of description of the present disclosure

and simplified description, rather than indicating or implying that the referred device or element must have a specific orientation, and is constructed and operated in a specific orientation, so the terms describing the working position relationships in the accompanying drawings are only used for exemplary description and cannot be understood as a limitation to the present disclosure, and for those who skilled in the art, specific meanings of above terms may be understood according to specific situations.

[0099] The above are only preferred embodiments of the present disclosure and are not intended to limit the present disclosure, and any modification, equivalent replacement and improvement made within a spirit and a principle of the present disclosure shall fall within a protection scope of the present disclosure.

## Claims

1. A sealing device for sealing a capsule product, the capsule product comprising a sealing piece configured to seal the capsule product, comprising: a base; and an upper cover; wherein the base defines an accommodating groove having an upper opening, and the base further comprises a positioning baffle disposed at one side of the accommodating groove; wherein the positioning baffle protrudes upwards from a side wall of the base and is configured to allow an outermost edge of the sealing piece along a thickness of the sealing piece to abut against an inner sidewall of the positioning baffle facing towards the accommodating groove, wherein the sealing piece is operative to be placed substantially perpendicular to the positioning baffle and abut against the inner sidewall of the positioning baffle for positional alignment; and wherein the upper cover is detachably connected to the base, a pressing portion is disposed on a lower surface of the upper cover.
2. The sealing device for the capsule product according to claim 1, wherein an annular protrusion is disposed on an upper surface of the base and is annularly disposed at an outer periphery of the upper opening of the accommodating groove.
3. The sealing device for the capsule product according to claim 2, wherein a positioning concave portion is further defined on the lower surface of the upper cover, the positioning baffle is at least partially inserted into the positioning concave portion slides in the positioning concave portion.
4. The sealing device for the capsule product according to claim 1, wherein the capsule product comprises a capsule shell and the sealing piece, the capsule shell comprises a cup body having a receiving groove, a flange connected to a top portion of the cup body and extending in a radial direction, and a crimping edge connected to an outer periphery of the flange, the sealing piece is attached to the flange and is configured to seal the receiving groove; and an annular limiting portion is further disposed on the lower surface of the upper cover and is annularly disposed at an outer periphery of the pressing portion, an annular recess is formed between the annular limiting portion and the pressing portion, the cup body is disposed in the accommodating groove, the flange is clamped between the pressing portion and the annular protrusion, the crimping edge is inserted into the annular recess, and an outer periphery of the crimping edge abuts against an inner wall surface of the annular limiting portion.
5. The sealing device for the capsule product according to claim 4, wherein the upper cover comprises a cover body and a top cover, the pressing portion is disposed on a bottom portion of the cover body, the pressing portion is of an annular structure matched with the annular protrusion, the cover body defines a through hole at an inner side of the pressing portion, the top cover is detachably connected to a top portion of the cover body for covering or exposing the through hole; and when the cover body is disposed on the base and the top cover is disconnected to the top portion of the cover body, a powdered product is placed into the receiving groove through the through hole.
6. The sealing device for the capsule product according to claim 4, wherein a protruding structure is disposed on an upper surface of the upper cover, a shape of the protruding structure is matched

with a shape of the upper opening of the accommodating groove, and the protruding structure is configured to flatten a powered product in the receiving groove.

7. The sealing device for the capsule product according to claim 2, wherein in a radial direction, a distance between of an inner wall surface of the positioning baffle and an outer wall surface of the annular protrusion is not less than 0.1 mm and is not greater than 5 mm.

8. The sealing device for the capsule product according to claim 2, wherein in a radial direction, a dimension of the annular protrusion is not less than 0.2 mm and not greater than 5 mm; and/or in a vertical direction, the dimension of the annular protrusion is not less than 0.1 mm and not greater than 10 mm.

9. The sealing device for the capsule product according to claim 1, wherein the pressing portion is a planar structure extending in a horizontal direction.

10. The sealing device for the capsule product according to claim 1, wherein the base defines a positioning groove at an outer periphery thereof, a positioning protruding portion is disposed at an outer periphery of the upper cover, the positioning groove is configured to limit the positioning protruding portion and is matched with the positioning protruding portion, so as to prevent the upper cover and the base from being separated from each other in a radial direction.

11. The sealing device for the capsule product according to claim 1, wherein the positioning baffle is of an arc-shaped structure, the base has a circle passing through the positioning baffle in a circumferential direction thereof, an arc length occupied by the positioning baffle on the circle does not exceed a circumference of the circle and is not less than one quarter of the circumference of the circle.

12. The sealing device for the capsule product according to claim 1, wherein an anti-slip pad is disposed on a lower surface of the base.

13. The sealing device for the capsule product according to claim 1, wherein the capsule product comprises a capsule shell and a sealing piece, the capsule shell comprises a cup body having a receiving groove, a flange connected to a top portion of the cup body and extending in a radial direction, and a crimping edge connected to an outer periphery of the flange, the sealing piece is attached to the flange and is configured to seal the receiving groove; the sealing device for the capsule product further comprises a cutting structure, the cutting structure comprises a mounting base and a blade assembly, a limiting groove is defined on a first side of the mounting base, the blade assembly comprises a fixing base and a blade fixed on the fixing base, the mounting base comprises a limiting hole penetrating through two opposite sides thereof, and the limiting hole is defined in the limiting groove, the blade partially extends out from the limiting hole; and an outer periphery of the capsule shell abuts against a groove side wall of the limiting groove, the capsule shell rotates in the limiting groove, the blade is configured to cut a part of the sealing piece when the capsule shell rotates with respect to the mounting base, wherein the part of the sealing piece is disposed at an inner side of the flange.

14. The sealing device for the capsule product according to claim 13, wherein an included angle is formed between the blade and a horizontal direction, the blade partially extends out from the limiting hole and is inclined in a direction away from an edge of the capsule shell.

15. The sealing device for the capsule product according to claim 14, wherein the included angle between the blade and the horizontal direction is  $\alpha$ , the  $\alpha$  satisfies a relationship as follows:  $55^{\circ} \leq \alpha \leq 95^{\circ}$ .

16. The sealing device for the capsule product according to claim 13, wherein the fixing base comprises a first base body and a second base body, the first base body is detachably connected to the second base body, the first base body comprises a first mounting surface, the first mounting surface is inclined, the second base body includes a second mounting surface, the second mounting surface is inclined, the blade is clamped between the first mounting surface and the second mounting, so as to form an included angle with a horizontal direction.

17. The sealing device for the capsule product according to claim 13, wherein a first mounting

groove and a second mounting groove are defined on a second side of the mounting base facing away from the limiting groove, the first mounting groove is communicated with the limiting hole, the blade assembly is fixed at the first mounting groove in a use state, and the blade assembly is accommodated in the second mounting groove in an idle state.

**18.** The sealing device for the capsule product according to claim 17, wherein the mounting base comprises a mounting portion, the mounting portion defines the first mounting groove, the mounting portion comprises a hook, the fixing base defines a clamping groove; and when the fixing base is fixed at the first mounting groove, the hook is snap-fitted to the clamping groove.

**19.** An operation method of a sealing device for sealing a capsule product, wherein the capsule product comprises a capsule shell and a powdered product contained in the capsule shell, wherein the capsule shell comprises a cup body and a sealing piece, wherein the cup body defines a receiving groove and comprises a flange disposed at a top portion thereof, the flange extending in a radial direction, and wherein the sealing piece is configured to be bonded to the flange so as to seal the receiving groove; wherein the sealing device comprises a base and an upper cover, wherein the base defines an accommodating groove having an upper opening, and wherein the base comprises a positioning baffle disposed at one side of the accommodating groove; wherein the positioning baffle protrudes upwards from a side wall of the base and is configured to allow an outermost edge of the sealing piece along a thickness of the sealing piece to abut against an inner sidewall of the positioning baffle facing towards the accommodating groove; wherein the sealing piece is operative to be placed substantially perpendicular to the positioning baffle and abut against the inner sidewall of the positioning baffle for positional alignment; wherein there is disposed an annular protrusion on an upper surface of the base and annularly disposed at an outer periphery of the upper opening of the accommodating groove, the accommodating groove being configured to accommodate the cup body, wherein the annular protrusion is configured to support the flange; wherein the upper cover is detachably connected to the base, and there is disposed a pressing portion on a lower surface of the upper cover wherein the operation method comprises: placing the cup body of the capsule shell of the capsule product in the accommodating groove of the base of the sealing device, allowing the flange of the cup body to be supported on the annular protrusion of the base; placing the powdered product into the receiving groove of the cup body; abutting the outermost edge of the sealing piece of the capsule shell along the thickness of the sealing piece against the inner sidewall of the positioning baffle of the base facing towards the accommodating groove in the base, and attaching the sealing piece to the flange; and covering the base by the upper cover of the sealing device of the capsule product, so as to seal and attach the sealing piece to the flange.

**20.** An operation method of a sealing device for a capsule product, comprising: placing a cup body of a capsule shell of the capsule product in an accommodating groove of a base of the sealing device for the capsule product, supporting a flange of the cup body on an annular protrusion of the base; placing a powdered product into a receiving groove of the cup body, or covering the base by a cover body of an upper cover of the sealing device for the capsule product and placing the powdered product into the receiving groove through a through hole of the cover body; removing the cover body, abutting an outer wall surface of a sealing piece of the capsule shell against an inner wall of a positioning baffle of the base, and attaching the sealing piece to the flange; and covering the base by the upper cover of the sealing device of the capsule product, so as to seal and attach the sealing piece to the flange; wherein the capsule product comprises the capsule shell and the powdered product contained in the capsule shell, the capsule shell comprises the cup body and the sealing piece, the cup body defines the receiving groove and comprises the flange at a top portion thereof, the flange extends in a radial direction, the sealing piece is attached to the flange and is configured to seal the receiving groove; the sealing device for the capsule product comprises the base and the upper cover; the base defines the accommodating groove, the accommodating groove has an upper opening, the annular protrusion is disposed on an upper surface of the base and is annularly disposed at an outer periphery of the upper opening of the accommodating groove, the

accommodating groove is configured to accommodate the cup body, the annular protrusion is configured to support the flange, the base comprises the positioning baffle disposed at one side of the accommodating groove, and the positioning baffle is configured to limit the sealing piece; and the upper cover comprises the cover body and a top cover, the cover body is detachably connected to the base, and a pressing portion is disposed on a bottom portion of the cover body, the cover body defines the through hole at an inner side of the pressing portion, the top cover is detachably connected to a top portion of the cover body for covering or exposing the through hole.

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